Clip Compass: Mapping CLIP Embedding Space through Perturbation

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What paper/papers did you choose?
We would like to do our project on Explainable Mapper: Charting LLM Embedding Spaces Using Perturbation-Based Explanation and Verification Agents.

What is the problem that you will be investigating? Why is it interesting?
We will be exploring CLIP image embedding spaces, investigating whether we can generate meaningful explanations of how images are grouped within these spaces. For example, we want to understand whether clusters are formed based on object identity, visual attributes (such as color), or other semantic relationships. While CLIP maps images and text into a shared embedding space, understanding what determines the structure of these embeddings remains challenging and results in issues like asking for a blue car but getting other colors too. Being able to identify and verify these relationships could help us better understand vision-language models and potentially improve downstream applications such as image retrieval.

What data will you use?
COCO images and object annotations, along with corresponding textual descriptions from the Localized Narratives dataset: https://google.github.io/Localized-narratives/.

What method/model will you use? Are there any existing implementations? If so, in which libraries are they written?
We will use a pretrained CLIP model to generate image embeddings and KeplerMapper, as used in the original paper, to construct a Mapper graph representing the embedding space. We will then create an explanation agent using OpenAI to analyze images and their associated descriptions within each cluster and generate hypotheses about the semantic properties that distinguish them. To verify these explanations, we will extend the original perturbation-based approach to work with images. Specifically, we will use COCO object annotations to mask relevant image regions, recompute their CLIP embeddings, and observe how their embedding similarities and positions change. We will compare these changes against the predictions made by the explanation agent.

How will you evaluate the results?
We plan to evaluate the effectiveness of our explanations using three approaches:

1. Explanation accuracy: We will measure whether the generated explanations distinguish images belonging to a particular Mapper cluster from images in other clusters, using CLIP image-text similarity scores.
2. Perturbation-based verification: We will mask objects associated with the generated explanations and measure the resulting changes in CLIP embeddings. We will compare these changes against randomly masked regions of similar size to determine whether the explanations identify meaningful visual features.

3. Comparison against a baseline: We will compare explanations generated using our verification approach against explanations generated by an LLM without verification, to determine whether the verification process improves explanation reliability.

What will be the innovation in your project?

While the original Explainable Mapper paper focuses on explaining text-based embedding spaces, we propose extending its approach to CLIP's multimodal image-text embedding space. Our main innovation will be developing an image-grounded verification mechanism that uses controlled visual perturbations, rather than relying only on text-based perturbations, to determine whether generated explanations accurately reflect the structure of CLIP's embeddings. Specifically, we will investigate whether modifying visual features identified by an explanation agent produces the predicted changes in CLIP's embedding space. This will allow us to evaluate not only whether a cluster explanation sounds reasonable, but whether it accurately captures the visual characteristics influencing the model's representations. We hope this extension will provide a more interpretable understanding of CLIP's embedding space and establish a foundation for improving downstream vision-language applications.